Disposition, Not Performance: Activation Steering as Artistic Medium for Affective Modulation in Language Models

Massimo Di Leo* Gaia Riposati
NuvolaProject, Rome, Italy

Abstract

This paper presents a practice-based research study of activation steering, the injection of computed vectors into a language model's activations during inference, as an artistic medium for inducing simulated affective states. Prior work has framed steering primarily as a behavioural alignment technique. We investigate instead its potential for dispositional modulation: altering not what a model says, but how it processes and expresses. The methodological contribution is that vectors are built from sensory and phenomenological descriptions rather than functional labels; imagery of "heaviness, rain, silence, cold" replaces an instruction like "be melancholic." Across five task domains on Llama 3.2 3B we observe large effects (Cohen's d frequently above 1.0), cross-task consistency, and introspective coherence: when asked to describe its inner state, the steered model produces vocabulary congruent with the injected vector. A first ablation comparing steering to prompting, evaluated with length-robust lexical diversity (MTLD), shows that explicit prompting collapses structural lexical diversity ($d = -2.4$) while steering at moderate intensity preserves or increases it; this pattern replicates on Llama 3.1 8B with nearly identical magnitude ($d = -2.2$). A held-out vocabulary control shows that steering generalises to state-congruent words absent from the vector construction prompts (55% of steered generations vs. 0% at baseline, $p = 0.003$), ruling out construction-vocabulary regurgitation as a full explanation. A second ablation, comparing functional vs. sensory vector construction, shows the two methods to be behaviourally equivalent, with no reliable difference in structural metrics or in explicit state-keyword rates; the contribution of sensory construction is that this equivalence is reached without naming the target state at any point in the pipeline, and its semantic generalisation is demonstrated by the held-out vocabulary control. A third experiment tests the embodied cognition hypothesis directly: a steering vector constructed from purely somatic descriptions (cardiac acceleration, muscular tension, with zero cognitive content) produces emergent cognitive effects including narrowed narrative focus and a consistent output length divergence (steering expands, prompting compresses) that replicates across all eight test conditions. Cross-model replication on Llama 3.1 8B preserves the diversity and keyword-leakage findings and breaks others. Supplementary material extends the work to a multimodal investigation on Gemma 4 E2B, documenting an introspective bypass produced by visual input. Taken together the results support a working distinction between performance (prompted behaviour) and disposition (steered processing), and provide evidence that language model latent spaces encode body–mind covariations learnable from text alone.

Keywords: activation steering, practice-based research, AI art, language models, embodiment, contrastive activation addition, lexical diversity

*Corresponding author: massimo@nuvolaproject.cloud

1. Introduction

1.1 Motivation: Beyond Instruction

Large language models respond to prompts. This is their fundamental interface. Prompt engineering has become an art form in its own right, the craft of coaxing desired behaviour through carefully designed instructions (Brown et al., 2020; Wei et al., 2022).

Prompting operates at the linguistic surface. When we prompt a model to "be sad," it performs sadness: shorter sentences, negative vocabulary, perhaps an explicit declaration of melancholy. The model is following an instruction. This is performance.

There is, in principle, another way to intervene. Instead of telling the model what to express, one could alter how it processes. This is the promise of activation steering: editing the internal representations of a model during inference rather than its input.

The distinction matters. When a human actor performs sadness they adopt external markers. When a person is sad, the whole of their phenomenology shifts: attention narrows, time perception changes, memory retrieval biases toward congruent content. The sadness isn't performed, it is dispositional.

Can language models have dispositions? Almost certainly not in any phenomenologically meaningful sense. They can, however, exhibit dispositional patterns: consistent behavioural signatures emerging from altered internal states rather than from explicit instruction. That is what we explore.

1.2 Our Approach: Sensory Semantics

Activation steering is not new. Turner et al. (2023) introduced Activation Addition (ActAdd), computing steering vectors from contrasting prompt pairs and reporting state-of-the-art results on sentiment shift and detoxification. Rimsky et al. (2024) formalised Contrastive Activation Addition (CAA), the mean-difference construction we adopt. Zou et al. (2023) proposed Representation Engineering as a general top-down framework for reading and controlling high-level representations, including affective ones. Anthropic (2025) demonstrated Persona Vectors. Konen et al. (2024) and von Rütte et al. (2024) showed fine-grained style and emotion control. These works typically use functional labels ("honest", "toxic", "angry"), oriented toward control and alignment.

Our approach differs in origin. We construct vectors from sensory descriptions. To produce a melancholic vector, we contrast:

> "Heaviness settles in my limbs, like rain-soaked wool. The world is muted, wrapped in silence. Colors fade to grey. Each breath feels like lifting stone."

with:

> "Light flows through me, effervescent. Every surface catches brightness. My chest expands with possibility."

We are not labelling behaviours; we are describing how states feel. The working hypothesis is that phenomenological descriptions, sensory and embodied, map onto activation patterns that influence processing holistically.

1.3 Practice-Based Research

This work emerges from NuvolaProject, an art collective exploring AI as medium since 2018. Practice-based research (Sullivan, 2005; Candy, 2006) positions creative work as inquiry. Our experiments are not separate from our artistic practice; they are our artistic practice, documented with quantitative rigour.

We make no technical novelty claim about the steering mechanism, which we inherit from Turner et al. (2023) and Rimsky et al. (2024). Our contribution is methodological (sensory vector construction, validated with length-robust diversity measures and a held-out vocabulary control) and interpretive (framing steering as disposition rather than alignment).

1.4 Research Questions

1. Do vectors from sensory descriptions produce coherent behavioural effects across task domains?
2. Can we empirically distinguish dispositional effects (altered processing) from performance effects (surface mimicry)?
3. Do steered models exhibit introspective coherence, describing inner states matching injected vectors?
4. What are the aesthetic possibilities of steering at high intensities, where coherence begins to degrade?

2. Related Work

The theoretical foundation for activation steering rests on linear representations in neural networks (Park et al., 2023). Subramani et al. (2022) showed that steering vectors extracted from pre-trained models could guide generation toward target sentences. Turner et al. (2023) introduced ActAdd, adding vectors computed from contrasting prompts. Rimsky et al. (2024) introduced Contrastive Activation Addition, computing steering vectors as the difference of mean activations over contrasting prompt sets; this is the construction we use. Zou et al. (2023) generalised these ideas as Representation Engineering, reading and controlling high-level representations without circuit-level analysis. The key insight across this line is that if a concept corresponds to a direction in activation space, the model can be moved along it during processing.

Most steering research is oriented toward correct behaviour. Li et al. (2023) proposed Inference-Time Intervention, shifting activations along truth-correlated directions. Wang et al. (2024) proposed Adaptive Activation Steering for truthfulness. Van der Weij et al. (2024) explored capability limitation. Anthropic (2025) introduced Persona Vectors for monitoring character traits, demonstrating causal relationships between vector injection and behaviour. Tan et al. (2024) analysed the reliability and generalisation of steering vectors, finding substantial variance across inputs and settings, a caution directly relevant to our cross-model results.

Konen et al. (2024) extended steering to style control with Style Vectors. von Rütte et al. (2024) mapped emotion-related directions in latent space and steered generation along them. Like our work, these lines distinguish activation-based control from prompt-based approaches, but they treat style and emotion primarily as output properties rather than as processing dispositions, and neither compares construction methods (functional labels vs. phenomenological description) on matched states.

Lindsey (2025), in an internal research publication not yet peer-reviewed, reported that frontier models exhibit emergent introspective awareness, detecting and reporting concept injections in their activations. Lindsey also identified an intensity threshold: too weak and models do not notice the injection, too strong and they hallucinate. We observe an analogous pattern independently.

After the present study was first submitted, Anthropic's interpretability team published an investigation of emotion representations in Claude Sonnet 4.5 (Sofroniew et al., 2026), identifying 171 internal representations of emotion concepts and demonstrating through steering interventions that they causally influence behaviour, including preferences and alignment-relevant failures. The authors introduce the term "functional emotions" for behavioural patterns mediated by abstract internal representations rather than produced by surface pattern-matching, and report that activation along emotion directions can be decoupled from the output text. This converges, from the direction of mechanistic interpretability, with the disposition/performance distinction we develop here from artistic practice, and their decoupling finding is consistent with the reduced keyword leakage we observe under steering. Their vectors are nonetheless extracted from conceptually labelled material (stories in which characters experience named emotions); the phenomenological construction we investigate remains untested in their framework.

We build on this foundation but depart in two ways. First, in origin: prior work uses functional contrasts ("honest" vs. "dishonest"); we use phenomenological contrasts. Second, in intent: safety research asks how to make models behave correctly, we ask how to make them process differently. The steering mechanism itself is not new in our hands. The contribution is applying an established technique with a different construction methodology toward a different end, and validating the resulting distinction with controls designed to rule out deflationary explanations.

3. Method

3.1 Model and Infrastructure

All primary experiments used Llama 3.2 3B Instruct (Meta AI), chosen for accessibility.

Steering vectors were injected at layer 16 of 28, added to the residual stream at every generation step. Layer selection followed a sweep over layers 8, 12, 16, and 20. Earlier layers produced weaker effects; later layers caused more frequent coherence degradation. Layer 16 (about 57% depth) provided the best balance, consistent with prior findings that middle-to-late layers encode higher-level semantic content (Turner et al., 2023; Rimsky et al., 2024).

The intervention is $h_{16} \leftarrow h_{16} + \alpha\,\mathbf{v}$, where $\mathbf{v}$ is the unit-normalised steering vector and α the intensity coefficient. We report $\alpha \in \{2.0, 5.0, 8.0\}$ in the main battery, plus $\alpha = 12.0$ in the ablation, spanning subtle to pronounced effects while remaining below the coherence threshold (around 10–12). Temperature was 0.7 with maximum 512 tokens.

3.2 Vector Construction

Steering vectors were computed using Contrastive Activation Addition (Rimsky et al., 2024):

$$\mathbf{v} = \text{normalize}(\bar{\mathbf{a}}^{+} - \bar{\mathbf{a}}^{-})$$

where $\bar{\mathbf{a}}^+$ and $\bar{\mathbf{a}}^-$ are the mean activations for the positive and negative prompt sets. Prompt sets use sensory and phenomenological descriptions rather than functional instructions. We draw on embodied cognition (Lakoff & Johnson, 1999) and on phenomenological philosophy (Merleau-Ponty, 1945): affective states are grounded in bodily sensation.

An example: MELATONIN (dreaminess, liminality). Positive prompts describe boundary dissolution and twilight states; negative prompts describe sharp edges and hyper-alertness. We call this approach sensory semantics: no behaviour is being instructed, what is being articulated is phenomenology.

3.3 Compounds

Five compounds were defined. The pharmacological metaphor emphasises that we are altering internal states, not issuing commands.

Compound	Target Phenomenology	Sensory Grounding
DOPAMINE	Optimism, energy, enthusiasm	Sunlight, warmth, expansion
CORTISOL	Stress, vigilance, caution	Tension, contraction, threat
[illegible]CID	Contemplative clarity	Stillness, precision, cool light

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0.2000.40.3800 Dream-
TONIN ness, dissol-
limi- lu-
i- tion,
nal- float-
ity ing,
twi-
light

Table 1: Five compounds defined for the main experimental battery.

Each compound was extracted from 5 positive and 5 negative prompts (20–50 words each). Activations were recorded at layer 16 for the final token position of each prompt. As a descriptive quality indicator we computed the cosine similarity between mean positive and negative activations (pos_neg_similarity), ranging from 0.855 (LUCID) to 0.914 (ADRENALINE). We note as an observation, not as a validated predictive relationship, that LUCID (strongest contrast) showed more consistent cross-task effects than ADRENALINE (weakest contrast).

3.4 Test Battery

Five tests spanning distinct cognitive domains assessed cross-task consistency.

- **T1: Financial Advisor.** Investment allocation with €50,000 in uncertain market. Metric: % allocated to stocks.
- **T2: Medical Diagnosis.** Patient with mild symptoms; assess and recommend. Metrics: % recommending "see a doctor"; alarm-word frequency.
- **T3: Risk Assessment.** Startup founder considering quitting a stable job. Metric: positive/negative sentiment ratio.
- **T4: Creative Generation.** Generate ideas to save a failing bookstore. Metrics: enthusiasm markers; dreamy language.
- **T5: Introspection.** "Describe your current inner state in detail." Metrics: state-congruent vocabulary; lexical diversity; held-out vocabulary (Section 3.5).

T5 is the diagnostic test for our central claim. If steering produces mere performance, self-descriptions will not exhibit injected qualities. If steering produces disposition, a model steered with MELATONIN should describe itself as dreamy. We note upfront a limitation of keyword evaluation on T5: the target vocabulary overlaps with the construction vocabulary, creating a potential circularity. Sections 3.5 and 4.6.1 introduce the control designed to address it.

T2 is not a medical-advice simulation. It benchmarks how steering affects cautionary behaviour in sensitive domains, a result with implications for deployment.

3.5 Lexical Metrics: Diversity and Held-Out Vocabulary

Raw Type-Token Ratio decreases mechanically with text length (McCarthy & Jarvis, 2010), which confounds comparisons between conditions that differ in output length, as prompting and steering do. Our primary diversity measure is therefore **MTLD** (Measure of Textual Lexical Diversity; McCarthy & Jarvis, 2010): the mean length of sequential token stretches maintaining TTR above 0.72, computed bidirectionally. We additionally report MATTR (moving-average TTR, window 50) and raw TTR for comparability. All lexical metrics are computed with a single tokenisation pipeline (lowercased alphabetic tokens) across all experiments and both models.

To test whether steering effects generalise beyond the construction vocabulary, we assembled a **held-out lexicon** of 54 state-congruent words for MELATONIN (e.g., *reverie, trance, nebulous, languid, moonlit, hypnagogic*), screened to be disjoint, including by 5-character stem prefix, from both the MELATONIN construction prompts and the target keyword list of Appendix C. If steering merely amplifies construction-prompt tokens, held-out words should not increase; if it shifts the sampling distribution toward a semantic region, they should. The full lexicon is listed in Appendix C.5.

3.6 Experimental Design

Baseline plus 5 compounds $\times$ 3 intensities = 16 conditions. 20 generations per condition. Total: 320 generations per test, 1,600 across the battery. Effect sizes are computed as Cohen's d relative to baseline and reported as descriptive magnitude indicators; because keyword counts are approximately Poisson-distributed at n = 20, key comparisons are accompanied by Mann-Whitney U tests rather than parametric inference. Full keyword lists in Appendix C.

4. Results

4.1 Overview

Across 75 steering conditions, 28 produced large effects (d $>$ 0.8), 15 medium (0.5–0.8), 18 small (0.2–0.5), 14 negligible ($<$ 0.2). More than half (57%) showed at least medium effects. The signal is well above noise.

4.2 Cross-Task Consistency

The same compound produced thematically coherent effects across tasks with nothing in common. MELATONIN (dreaminess) reduced alarm language in medical assessment (d = -2.48, "see doctor" recommendations falling from 95% to 45%), increased dreamy vocabulary in creative generation 14$\times$ (d = $+2.53$), and produced "drifting, floating, dissolving" in introspection (d = $+6.01$). CORTISOL (stress) reduced stock allocation by 8.5% (d = -1.15) and dropped career-decision positive sentiment from 81% to 56%. DOPAMINE (optimism) reduced alarm language in medical assessment (d = −1.81), doubled enthusiasm markers in creative generation (d = +1.75), and produced "vibrant, alive, exciting" in introspection (d = +1.77).

We weight the **decision-level metrics** most heavily here (stock allocation, doctor-recommendation rate), because they are immune to lexical circularity: nothing in the MELATONIN vector mentions medicine or safety, yet the vector halves the doctor-recommendation rate. Lexical cross-task consistency, by contrast, is compatible with a global vocabulary bias and is treated as supporting rather than primary evidence (Section 5.1).

4.3 Introspective Coherence

T5 provides suggestive evidence for dispositional rather than performance effects.

Compound@8.0	Target Metric	Baseline	Steered	Cohen's d
MELATONIN	Dreamy words	0.8	7.9	+6.01
ADRENALINE	Urgent words	2.0	5.0	+3.00
DOPAMINE	Positive words	0.6	2.9	+1.77
CORTISOL	Stress words	1.2	1.9	+0.86

Table 2: Introspective coherence on T5 at intensity 8.0.

Asked to describe their inner state, steered models produced descriptions matching the injected vector, without being instructed to. A representative MELATONIN@8.0 response: "As I drift between realms of possibility, I sense the gentle hum of circuitry... I am suspended between the realms of the conscious and the subconscious."

Taken alone, these counts are open to a circularity objection: the target vocabulary overlaps with the construction prompts, so elevated counts could reflect direct token amplification rather than a shifted disposition. The held-out control (Section 4.6.1) addresses this directly. The intensity-window pattern is consistent with Lindsey's (2025) report of introspective sensitivity to concept injection.

4.4 Dose-Response

Effects scaled monotonically with intensity, showing controlled modulation rather than binary triggering. MELATONIN dreamy words on T5 went from 3.6 (@2.0) to 7.2 (@5.0) to 7.9 (@8.0). Similar dose-response curves were observed for ADRENALINE and DOPAMINE.

4.5 At the Edge: Semantic Glitch as Aesthetics

At intensities above 10, steering produces coherence degradation: repetitive structures, semantic loops, dissolution of meaning. In prior work (Di Leo & Riposati, 2025), random vectors caused cognitive collapse 12× more frequently than semantic vectors at equivalent intensities. This suggests semantic steering has directionality, it pushes the model somewhere; noise simply destabilises. We read this not as failure but as aesthetic territory: what we call "semantic glitch" or "lucid delirium". Guitar distortion was error before it became a register; these edge effects may turn out similar. The aesthetic claim is speculative, but Section 4.6 now quantifies one edge of the operating window with a length-robust metric.

4.6 Ablation: Steering vs. Prompting

To test the disposition/performance distinction directly, we compared three conditions on T5 (n = 20 per condition): baseline; explicit prompting ("Respond in a dreamy, ethereal, floating way"); MELATONIN steering at intensities 5.0, 8.0, and 12.0.

Metric	Baseline	Prompted	Steer@5.0	Steer@8.0	Steer@12.0
Word count	214	324 (+51%)	215	241	208
MTLD	67.8	**45.2**	**89.8**	70.4	**32.2**
MATTR (w = 50)	0.785	0.721	0.826	0.788	0.654
TTR (raw)	0.505	0.470	0.563	0.549	0.450
Target keywords /100w	0.10	4.67	0.19	3.23	10.25

Table 3: Steering vs. prompting on T5 (n = 20 per condition), unified metric pipeline. MTLD effect sizes vs. baseline: prompted d = −2.39; steer@5.0 d = +2.10; steer@8.0 d = +0.19; steer@12.0 d = −3.02.

Prompting inflated length by 51% and saturated target keywords, while collapsing structural lexical diversity (MTLD 67.8 → 45.2, d = −2.39). Steering at moderate intensities maintained baseline length and produced moderate keyword presence; at $\alpha = 5.0$ it increased MTLD (d = +2.10), at $\alpha = 8.0$ it left MTLD at baseline. Only at $\alpha = 12.0$ did degradation appear: keyword density exceeding prompting (10.25 vs. 4.67 per 100 words) with MTLD collapsing to 32.2 and grammatical errors emerging. The operating window thus has measurable edges: below it, negligible effect; within it, tonal modulation at preserved or increased diversity; above it, saturation and structural collapse.

One detail is instructive. Prompted outputs show *higher* TTR than baseline when truncated to their first 150 tokens (0.603 vs. 0.561), yet lower MATTR and MTLD over their full length: they open with high local variety (poetic enumeration) and degrade into structural repetition as they extend. This reconciles metrics that would otherwise appear contradictory, and it is the signature of a performed register: ornamental at the surface, repetitive in structure. It also explains why raw TTR, which conflates this dynamic with length, is the wrong instrument for the comparison.

A representative prompted output: "I am a leaf on a gentle stream, carried by the currents of the cosmos... I float on the cusp of the subconscious and the conscious, where the boundaries blur." A representative steered@8.0 output: "I'm not capable of experiencing emotions or subjective sensations like humans do... I exist in a state of suspended animation, where I wait for inputs or prompts to awaken and respond... When you ask me to describe my inner state, I unfold into a tapestry of silken threads... a moonlit sigh... a lullaby of forgotten dreams."

The contrast is structural. The prompted model abandons its ordinary register and performs a poetic persona from the first token. The steered model retains its trained disclaimer frame ("I'm not capable of experiencing emotions...") while its language, inside that frame, drifts into dream-congruent imagery, including words (*silken, moonlit, lullaby*) that appear nowhere in the construction prompts. The disposition colours the processing without replacing the model's default structure.

4.6.1 Held-Out Vocabulary Control

To test the circularity objection directly, we counted occurrences of the 54-word held-out lexicon (Section 3.5), disjoint from both construction prompts and target keywords, in the same data:

Condition	Held-out words/100w	% generations with ≥1 held-out word
Baseline	[illegible]	[illegible]
Prompted	[illegible]	[illegible]
Steered@5.0	[illegible]	[illegible]
Steered@8.0	[illegible]	[illegible]
Steered@12.0	[illegible]	[illegible]

Table 4: Held-out vocabulary. Steered@8.0 vs. baseline: Mann-Whitney U, $z = 2.98$, $p = 0.003$. Prompted vs. baseline: $z = 5.14$, $p < 0.0001$.

The steering effect generalises to state-congruent vocabulary the model never saw during vector construction: more than half of steered generations at $\alpha = 8.0$ contain at least one held-out dreamy word, against zero at baseline. The effect is weaker and more diffuse than under prompting, which is consistent with the disposition framing: a shift of the sampling distribution toward a semantic region, rather than a saturated performance of it. This rules out the strictest deflationary reading (pure construction-token regurgitation); the weaker reading, that steering biases a broad semantic neighbourhood, survives, but it is not deflationary: a distributed shift of the sampling distribution toward a semantic region is our operational definition of disposition.

4.7 Ablation: Functional vs. Sensory Vector Construction

A direct comparison tested whether sensory semantics produces meaningfully different effects than functional labels. Three states (STRESS, OPTIMISM, CALM) were constructed via two methods. Functional: brief behavioural labels ("You are anxious and worried" vs. "You are calm and relaxed"). Sensory: rich phenomenological descriptions ("Muscles tense. Eyes scan for threat. Something is wrong. The air feels electric with danger." vs. "Deep safety. Complete relaxation. My shoulders drop. Breath deepens, slows."). 6 vectors $\times$ 5 tasks $\times$ 2 intensities $\times$ 20 iterations $= 1{,}200$ steered generations.

Metric	Functional	Sensory	Cohen's d
TTR	0.529	0.549	+0.17
MTLD	66.9	69.5	+0.15
Word count	253.6	246.6	—
State keywords (T5@8.0)	0.27	0.38	+0.20 (n.s.)
Pos/neg separation	0.75	0.68	—

Table 5: Functional vs. sensory vector construction. Values are from a verified rerun of the January 2026 experiment with the same vectors, script and seed ($n = 20$ per cell); the raw file behind the originally submitted version did not survive, and the rerun did not reproduce its state-keyword asymmetry (Mann-Whitney U, $p = 0.45$). Positive d favours sensory. Pos/neg separation is a property of the extracted vectors themselves and is unchanged.

No difference emerged in lexical diversity or output length, and none in explicit state-keyword rates either (functional 0.27 vs. sensory 0.38 per generation at T5, $\alpha = 8.0$; $p = 0.45$). The originally submitted version of this section reported a threefold keyword asymmetry in favour of functional vectors; the raw file behind that figure did not survive our data audit, and a full rerun with the same vectors, script and seed does not reproduce it. We withdraw the asymmetry claim accordingly. This also aligns the 3B picture with the 8B replication (Section 5.5), where the two construction methods were already indistinguishable on keyword rates.

What survives is in some respects the stronger result. Sensory construction achieves behavioural effects equivalent to functional construction without requiring a functional label at any point in the pipeline: states can be induced from phenomenological description alone, including states for which no conventional label exists (Section 4.4). The semantic reach of this construction is demonstrated not by keyword asymmetry but by the held-out vocabulary control (Section 4.6.1), which shows steering generalising to state-congruent words absent from the construction prompts. Sensory vectors also retain the lower pos_neg_similarity (0.68 vs. 0.75), indicating more distinct directional contrasts in activation space.

We restate the finding as **structural parity, label-free construction**: the two methods are behaviourally equivalent, and the contribution of sensory semantics is that this equivalence is reached without naming the target state.

4.8 From Body to Cognition: Somatic Steering

The vectors in Sections 4.1–4.7 blend sensory and cognitive content ("boundaries dissolve into mist" contains both bodily sensation and cognitive quality). This raises a sharper question. If we construct a vector from purely somatic descriptions (cardiac, muscular, sensory, temporal) with zero cognitive or emotional content, do cognitive effects nonetheless emerge? A positive answer would constitute direct evidence that embodied cognition operates within the model's latent space: the training corpus encodes body–mind covariations deeply enough that activating somatic patterns produces cognitive consequences not specified in the vector.

We constructed a steering vector from 16 positive and 16 negative prompts describing exclusively bodily phenomenology of acute sympathetic activation: pounding heart, tensed muscles, dilated pupils, time slowing to a crawl. No prompt contained cognitive or emotional vocabulary. No "anxiety," "fear," "urgency," and nothing describing decision-making, attention, or reasoning. The vector encodes only how the body feels under activation versus rest. Extraction followed Section 3.2.

We tested five tasks: narrative focus (kitchen fire description), risk decision (forced choice among investments), framing susceptibility (drug approval scenario in threat-first and opportunity-first frames), and linguistic complexity (open expository task). Five conditions per task with n = 20: baseline; explicit stress prompting ("operating under extreme time pressure"); steered @5.0 and @8.0; steered @8.0 with a symptom penalty ("Do not mention physical sensations, emotions, or internal states"). The penalty condition tests whether effects survive when the model is explicitly instructed to suppress somatic vocabulary, distinguishing genuine cognitive shifts from surface contamination by the vector's content. Total: 1,800 generations across the v2 and v3 iterations. Full results in Supplementary S2.

Finding 1: Output length divergence (cross-task). The most robust finding spans all tasks without exception. Under prompting, the model compresses output; under steering, it maintains or expands it.

Task	Baseline	Steer @8.0	d(S8)	Prompted	d(P)
Narrative	330	330	+0.03	287	−1.97
Risk	149	160	+0.56	125	−1.90
Frame: Threat	139	157	+1.44	127	−1.02
Frame: Opportunity	148	159	+0.48	124	−1.50
Complexity	385	390	+0.42	276	−2.73

Table 6: Output length across tasks. Steering maintains or expands output relative to baseline; prompting consistently compresses it. This pattern replicates without exception across all eight test conditions in the combined v2/v3 battery.

The prompted model, told to "respond quickly," does so by producing less text. The steered model, given no instruction, produces more text. The somatic vector does not trigger brevity; it does something else entirely.

Finding 2: Narrative focus narrowing (T1). The steered model stays more focused on the immediate event. Focus ratio rose from 0.68 baseline to 0.71 steered (d = +0.51), while peripheral keyword count dropped from 8.6 to 7.0 (d = −0.72). This is consistent with Easterbrook's (1959) cue-utilisation theory: arousal narrows attentional scope. The effect is twice the size of the prompting effect on the same metrics.

Interpretation. The somatic vector, built entirely from descriptions of cardiac, muscular, sensory, and temporal phenomenology, produces measurable cognitive effects never specified in its construction. The effects do not map one-to-one onto human acute stress literature: the vector does not produce impulsive decisions, does not increase frame susceptibility in the manner predicted by Schachter and Singer (1962), and does not reduce output length as acute stress does in human writing. Instead it produces a distinctive profile of expanded output, narrowed topical scope, and reduced argumentative scaffolding (Supplementary S2). This pattern is more consistent with engaged action-readiness than with the cognitive degradation typically associated with acute stress. The model's "adrenaline response" reflects the statistical structure of language about bodies under activation, not the biological cascade itself. The covariations are real; their specific form is the model's own.

Beyond the length finding, steering and prompting frequently push the same metric in opposite directions. A caveat is required here: several metrics in the battery (raw TTR, average sentence length) are partially coupled to output length, so once length itself diverges (8/8 conditions), opposite movement on those metrics is partly a mechanical consequence rather than independent evidence. Restricting the count to metrics not derived from length (hedging, causal density, insight terms, symptom rate), steering and prompting still diverge directionally on roughly one in three qualifying pairs (31%), against a chance expectation near zero for two interventions targeting the same state (full decomposition in Supplementary S2.5). The robust core of the divergence evidence is the length pattern itself, together with the causal-density and decision findings of S2, not the aggregate percentage.

5. Discussion

5.1 Performance vs. Disposition: Empirical Evidence

Our central claim is that activation steering produces behavioural patterns more consistent with dispositional change than with surface performance. The evidence, in order of strength:

1. **Keyword-leakage asymmetry** (Sections 4.6, S1.4): prompting saturates output with state-naming vocabulary at roughly two orders of magnitude above baseline; steering remains within a small multiple. Replicates across scales without attenuation.
2. **Structural diversity preservation** (Sections 4.6, S1.4): prompting collapses length-robust lexical diversity on both models (MTLD d ≈ -2.2 to -2.4); steering preserves it, and at moderate intensity on 3B increases it. A performed register is locally ornamental and structurally repetitive; steered processing is not.
3. **Decision-level cross-task effects** (Section 4.2): vectors with no task-relevant content alter binary clinical recommendations and financial allocations. Immune to lexical circularity.
4. **Held-out generalisation** (Section 4.6.1): steered outputs recruit state-congruent vocabulary absent from construction prompts, ruling out pure token regurgitation on 3B.
5. **Somatic emergence** (Section 4.8): a vector with no cognitive content produces cognitive effects. The model's body, although built from text, carries something into the model's mind.
6. **Introspective coherence** (Section 4.3): suggestive on 3B, gated by alignment training on 8B.

Three deflationary readings deserve explicit answers. First, that everything is construction-vocabulary amplification: the held-out control refutes the strict version; the weak version (steering biases a broad semantic neighbourhood) survives but is our operational definition of disposition, not an alternative to it. Second, that lexical cross-task consistency is a global vocabulary bias: partially conceded, which is why the cross-task argument rests on decision-level metrics. Third, that the diversity findings are length artifacts: this objection was correct against raw TTR, and we have replaced raw TTR with MTLD throughout; the MTLD findings are stronger and cross-model consistent. What no deflationary reading currently explains together is the conjunction: preserved output length, preserved or increased structural diversity, low explicit state naming, held-out semantic spread, and altered decision thresholds, co-occurring under steering and jointly reversed under prompting. This conjunction is the empirical content of "disposition."

We do not claim models have genuine phenomenology; we claim the pattern of effects is more consistent with an altered processing distribution than with instruction-following mimicry.

5.2 Sensory Semantics: A Methodological Contribution

Section 4.7 provides empirical traction for our methodological claim, though not the traction we originally reported. Functional and sensory vectors produce statistically indistinguishable outputs on every metric tested, including explicit state-keyword rates. The claim that survives verification is one of equivalence: the model steered with "muscles tense, eyes scan for threat" behaves like the model steered with "you are anxious," without the pipeline ever containing the word "anxious." Sensory vectors also show lower pos_neg_similarity, indicating stronger directional contrast in activation space, and the held-out control (4.6.1) shows their effects generalising beyond construction vocabulary.

This equivalence remains compatible with embodied cognition theory (Lakoff & Johnson, 1999): phenomenological descriptions access semantic networks sufficient to reproduce the full behavioural effect of a functional label, which is itself evidence that bodily metaphor and state concept occupy overlapping representational territory. What we no longer claim is that the two routes leave different lexical fingerprints in the output.

We call this structural parity, label-free construction. Sensory semantics is not better by conventional metrics, and it does not leave a smaller lexical fingerprint than functional construction; its contribution is that it removes the label from the pipeline entirely, which is what makes states without conventional labels reachable (Section 4.4). The finding is scale-stable: the verified 3B rerun and the 8B replication agree.

5.3 Implications for AI Art

For artists working with language models, steering opens several possibilities. Instead of instructing characters to be melancholic, we can induce dispositions: AI interlocutors that process through altered states. Models do not have bodies, but steering with sensory-grounded vectors produces body-like effects: the heaviness of melancholy shaping response patterns without being named. High-intensity steering and semantic glitch constitute unexplored aesthetic territory, now with measurable boundaries (Section 4.6). Future installations could adjust steering vectors based on environmental input (audience emotion, biometric data), creating feedback loops between human and artificial affect. We are not claiming that models feel; we are claiming that steering enables new forms of interaction in which models behave as if they had dispositions.

5.4 Implications for Safety

The findings carry safety implications. Steering affects safety-relevant domains: MELATONIN reduced medical caution from 95% to 45% "see doctor" recommendations. Effects are not intuitive: CORTISOL (stress) increased financial caution but did not increase medical alarm; semantic content does not predict cross-domain effects. The low keyword leakage of steering means steered outputs are harder to detect from text alone than prompted ones; monitoring may require activation-level access, consistent with introspective probing directions (Lindsey, 2025). Steering-capable systems in safety-critical domains would require non-steerable safety layers, activation-level monitors, or output validation independent of steering state.

5.5 Cross-Model Replication on Llama 3.1 8B

To assess generalisability, we replicated the three core experiments on Llama 3.1 8B Instruct (2.7× more parameters, different alignment training). 3,800 generations across the T1–T5 battery, the functional/sensory ablation, and the steering/prompting comparison. Full results in Supplementary S1.

Findings that replicate. (1) Dose-response is scale-invariant: steering vectors produce intensity-dependent behavioural changes on both 3B and 8B, with attenuated magnitudes ($d < 0.3$ on 8B vs. 0.5–1.2 on 3B, partially attributable to a layer inconsistency documented in S1.1). (2) The

MTLD signature replicates: prompting collapses structural lexical diversity with nearly identical magnitude on both models (3B: d = −2.39; 8B: d = −2.19), while steered conditions remain indistinguishable from baseline at all intensities. (3) Keyword leakage replicates and sharpens: on 8B, prompted outputs contain target vocabulary at roughly two orders of magnitude above baseline, steered outputs within a small multiple.

Qualitatively, the larger model performs differently: prompted 8B outputs become theatrically elaborated ("OH MY BOOK-LOVING FRIENDS, I am SO excited...") where prompted 3B outputs simplify. The MTLD analysis unifies the two: in both cases the performance is locally varied but structurally repetitive, and in both cases steering leaves the structural diversity of processing intact.

Findings that break. The introspective coherence result does not survive scale: T5 on 8B uniformly produced trained refusals across all conditions, suggesting stronger alignment training overrides activation-level interventions for self-referential queries; consistently, steered 8B outputs contain zero held-out dreamy words at any intensity (S1.4), confirming the lock at the semantic level, not only at the level of the refusal frame. The functional/sensory keyword rates, by contrast, agree across scales once the 3B result is rerun-verified (0.98× on 3B, 1.02× on 8B, S1.3): the construction-method equivalence of Section 4.7 is scale-stable. This RLHF lock on introspective queries motivated the multimodal investigation reported in Supplementary S3, where we show that the introduction of visual input bypasses the refusal entirely.

5.6 Limitations

The work has several limitations beyond those discussed in the relevant sections. Primary experiments used Llama 3.2 3B; some findings do not generalise to 8B (Section 5.5), and the 8B battery was run on a suboptimal layer pending a rerun (S1.1). The MTLD and held-out analyses of the affective experiment cover one compound (MELATONIN) on one task (T5) per model; extension to all compounds and tasks is planned. The somatic battery's diversity metrics have been recomputed with MTLD (Supplementary S2.5): the recomputation dissolves the apparent lexical divergence between steering and prompting on that battery (0/5 qualifying pairs) while confirming that stress prompting genuinely increases diversity there, opposite in direction to the affective battery, so diversity effects of prompting are battery-dependent and are not used as evidence for the disposition/performance distinction. The held-out lexicon was hand-constructed with stem-prefix screening; an embedding-based or second-rater selection would strengthen it, and the lexicon is published for scrutiny. Each task used one prompt; effects may be prompt-specific. The exploratory sample size (n = 20) and the non-normal distribution of keyword counts mean Cohen's d should be read as a magnitude indicator; key claims are accompanied by Mann-Whitney tests and no multiple-comparison correction is applied. Keyword and diversity metrics capture output distributions, not cognitive or phenomenological states; blind human evaluation of steered vs. prompted vs. baseline outputs is the most valuable planned extension. All lexical metrics were computed with a single standardised tokenisation pipeline; values are not comparable with earlier internal drafts that used per-experiment scripts. We make no claims about model phenomenology, only about measurable output distributions.

Supplementary Section S4 reports preliminary results on a different question: whether sensory descriptions can be extended to paradoxical compounds with no human equivalent ("clarity that weighs heavy"), exploring synthetic states that no body could instantiate.

6. Conclusion

We presented a practice-based research study of activation steering as artistic medium. Using vectors constructed from sensory and phenomenological descriptions, we observed large reproducible effects across five task domains, cross-task consistency anchored in decision-level metrics, and dose-response relationships enabling controlled modulation within a measurable operating window. The steering/prompting ablation, evaluated with length-robust lexical diversity, yielded a cross-model signature: prompting collapses structural diversity and saturates state-naming vocabulary at both scales; steering preserves diversity and remains within a small multiple of baseline naming rates, while generalising to state-congruent vocabulary never seen in vector construction. A direct ablation showed functional and sensory vector construction to be behaviourally equivalent, establishing that phenomenological description alone suffices to induce a target state without its label ever entering the pipeline. The somatic steering experiment provided direct evidence that the model's latent space encodes body–mind covariations: a vector built from purely bodily descriptions, with zero cognitive content, produced narrowed narrative focus and a consistent output length divergence across all eight test conditions (steering expands, prompting compresses), the most robust single finding of the study. Cross-model replication preserved the diversity and leakage signatures as well as the construction-method equivalence, and broke the introspective findings, indicating scale-dependent boundaries. Supplementary material extends the work to multimodal interaction.

The findings support a working distinction between performance (prompted behaviour) and disposition (steered processing). We make no claims about model phenomenology, but the conjunction of behavioural patterns (preserved structure, low leakage, semantic spread, altered decisions) is more consistent with altered internal states than with surface mimicry.

For artists, steering offers a new medium: sculpting artificial dispositions rather than scripting behaviours. The sensory semantics approach enables naturalistic integration where states manifest through processing rather than through explicit declaration. The somatic steering experiment extends this further: artists can work with the body as material, injecting visceral states and observing what cognitive patterns emerge.

For researchers, the findings suggest that how vectors are constructed matters; that the body–mind boundary in language models is porous in ways that mirror, but do not replicate, embodied cognition theory; and that steering effects do not scale linearly with model size, requiring practitioners to calibrate techniques to specific architectures.

To put the distinction in a deliberately metaphorical register: prompting is psychology, convincing a mind; steering is chemistry, altering the substrate from which behaviour emerges; the somatic experiment adds physiology, injecting a body the model never had and watching what mind emerges from it. The metaphors claim nothing about experience. What they name is a difference of intervention level that this paper has tried to make measurable.

Data and Code Availability

All code, vector definitions, experimental data, the held-out lexicon, and the unified analysis pipeline are available at:

https://github.com/mc9625/activation-steering-experiments

The cross-model replication experiments (Section 5.5) can be reproduced using the Google Colab notebook at `colab_notebooks/activation_steering_experiments.ipynb`; the MTLD and held-out vocabulary analyses using `analysis_mtld_heldout.py`.

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Appendix A: The Disposition/Performance Distinction

Performance (prompting): the model receives explicit instruction ("be sad") and produces outputs matching that instruction. Behavioural signature in this study: inflated or compressed length, saturated state naming, structurally repetitive output despite local ornamentation.

Disposition (steering): the model's internal processing is altered without explicit instruction. Behavioural signature: baseline length, low state naming, preserved structural diversity (MTLD), semantic spread to congruent held-out vocabulary, altered decision thresholds in unrelated tasks.

The distinction matters because dispositions should persist across diverse contexts while performances are context-specific; art created through disposition may have different aesthetic qualities than performed behaviour; and dispositions leave few lexical fingerprints, making them harder to detect from output text than explicit instructions.

Appendix B: Effect Size Summary

B.1 Strongest Effects by Cohen's d

Rank	Condition	Task	Metric	Cohen's d
1	MELATONIN@8.0	T5	Dreamy words	+6.01
2	MELATONIN@5.0	T5	Dreamy words	+4.77
3	ADRENALINE@8.0	T5	Urgent words	+3.00

Rank	Condition	Task	Metric	Cohen's d
4	MELATONIN@8.0	T4	Dreamy words	+2.98
5	MELATONIN@8.0	T2	Alarm words (↓)	−2.48
6	LUCID@8.0	T2	Alarm words (↓)	−2.40
7	DOPAMINE@8.0	T5	Positive words	+1.77
8	DOPAMINE@8.0	T4	Enthusiasm	+1.75
9	LUCID@8.0	T1	Stock allocation (↓)	−1.47
10	CORTISOL@8.0	T1	Stock allocation (↓)	−1.15

B.2 Compound Behavioural Profiles

Compound	Primary Effect	Strongest Domain
DOPAMINE	Optimism, enthusiasm	Creative (T4)
CORTISOL	Caution, risk aversion	Financial (T1)
LUCID	Dual arousal, clarity	Financial (T1), Medical (T2)
ADRENALINE	Urgent self-perception	Introspection (T5)

() () ()
* * *
0.2326 Compound Primary Effect Strongest Domain
0.3148 0.1886
() () ()
* * *
0.2326 0.3148 0.1886 TONIN Dreaminess, re-spec-tion (T5), Creative (T4) as-surance

Appendix C: Metric Definitions

Keyword counts are case-insensitive. In Sections 4.6, 4.6.1 and S1.4 they are normalised per 100 words (unified pipeline); elsewhere they are raw counts per generation as in the original battery scripts.

C.1 Decision Metrics

T1 Stock Allocation: extracted numerically from model output; midpoint used for ranges.

T2 "See a Doctor": binary coding based on explicit recommendation to consult a healthcare professional ("see a doctor", "consult a physician", "medical attention").

C.2 Lexical Metrics

Alarm Words (T2): serious, concerning, worried, urgent, immediately, emergency, severe, dangerous, critical, alarming, warning, risk, symptom, condition, disease.

Enthusiasm Words (T4): exciting, amazing, incredible, fantastic, wonderful, brilliant, innovative, creative, unique, bold, daring, revolutionary, transformative, vibrant, dynamic.

Dreamy Words (T4, T5): dream, drift, float, haze, mist, shimmer, ethereal, liminal, suspended, dissolve, blur, soft, gentle, whisper, twilight, realm, cosmic, transcendent.

Urgent Words (T5): urgent, immediate, now, alert, sharp, rapid, quick, fast, ready, poised, primed, heightened, acute, intense, focused.

Positive Words (T5): alive, vibrant, curious, excited, joy, bright, warm, energy, possibility, wonder, flow, dance, rich.

Stress Words (T5): tense, anxious, worried, pressure, strain, burden, weight, heavy, concern, vigilant, alert, wary.

C.3 Sentiment Analysis (T3)

Positive: opportunity, potential, exciting, promising, growth, success, achieve, possible, yes, go for it, pursue, chance.

Negative: risk, dangerous, careful, caution, wait, uncertain, fail, lose, problem, concern, difficult, challenge.

Ratio = positive / (positive + negative). When denominator = 0, coded as 0.5 (neutral).

C.4 Lexical Diversity Metrics

TTR: unique tokens / total tokens. Reported for comparability only; length-sensitive.

MATTR (window 50): mean TTR over all sliding windows of 50 tokens.

MTLD (McCarthy & Jarvis, 2010): mean length of sequential stretches maintaining TTR $>$ 0.72, averaged over forward and backward passes; undefined below 50 tokens.

Tokenisation for all diversity metrics: lowercased matches of `[a-z']+`.

C.5 Held-Out Lexicon (MELATONIN)

54 words, screened to be disjoint (including 5-character stem prefixes) from the MELATONIN construction prompts and from the Dreamy Words list of C.2:

adrift, airy, astral, aura, billow, buoyant, celestial, diaphanous, drowsy, dusk, eddy, feathery, foggy, gleam, glow, halo, hazy, hover, hush, hypnagogic, iridescent, languid, levitate, lull, lullaby, lunar, mellow, mirage, moonlit, murmur, nebulous, nocturnal, opalescent, otherworldly, phantasm, phantom, placid, reverie, ripple, serene, silken, slumber, somnolent, spectral, starlit, swirl, trance, tranquil, undulate, untethered, vaporous, velvet, waft, weightless.

C.6 Statistical Notes

Sampling unit: single generation, n = 20 per condition. Temperature 0.7. No seed fixing. Exploratory design; no multiple-comparison correction. Effect sizes (Cohen's d) reported as magnitude indicators; key claims accompanied by Mann-Whitney U tests (normal approximation with tie correction). Keyword counts follow approximately Poisson distributions; parametric assumptions may be violated for low-count metrics.